

Umer Rauf

AI & Software Engineer

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Profile

Computer Science Master's student with professional full-stack software engineering experience and a strong technical focus on applied machine learning and backend architecture. Proven ability to build scalable software solutions (Python, TypeScript, Node.js) and train deep learning models for complex data processing (PyTorch, Computer Vision). Adept at combining analytical problem-solving with modern cloud and data technologies to deliver robust, production-ready applications.

Education

04/2024 - 09/2027 **M.Sc. Computer Science (Expected)**

Universität Siegen, Germany

- Academic focus on Machine Learning, Artificial Intelligence, and Embedded Systems.

09/2018 - 11/2022 **B.Sc. Software Engineering**

University of Gujrat, Pakistan

Professional Experience

04/2023 - 01/2024 **Associate Software Engineer**

Allied Consultants, Lahore, Pakistan

- Developed and maintained data-driven software architectures and SaaS applications using Python, TypeScript, and Node.js.
- Designed complex relational databases (PostgreSQL) and managed efficient data workflows for high-throughput applications.
- Automated deployments and managed containerized environments using Docker and CI/CD pipelines.
- Diagnosed software defects, implemented analytical bug fixes, and collaborated seamlessly across technical disciplines in an agile environment.

02/2020 - 03/2023 **Junior Frontend Developer**

Hisons Tech, Gujrat, Pakistan

- Developed responsive user interfaces and data visualization dashboards to present complex system metrics clearly to end-users.

Selected AI & Software Projects

Machine Learning **U-Net Deep Learning Image Segmentation**

Python, PyTorch, Computer Vision

- Developed, trained, and evaluated a deep learning model to automatically classify and segment

complex spatial anomalies within visual datasets.

- Managed the end-to-end machine vision pipeline, including data normalization, image augmentation, and precise boundary prediction.
- Evaluated model reliability using statistical overlap metrics to ensure robust prediction capabilities.

AI Integration Track AI (Intelligent Data Processing)

Python, LLM APIs, Next.js

- Engineered an automated workflow that ingests unstructured inputs, performs AI-driven classification, and executes intelligent data structuring to a centralized dashboard.
- Integrated advanced AI models to extract key metrics and optimize analytical reporting processes.

Live: track-ai-one.vercel.app

Backend Serverless CRM Platform Architecture

React, Node.js, AWS Lambda, DynamoDB

- Architected and deployed a full-stack personal project utilizing serverless infrastructure to ensure high scalability and low latency.

Live: serverless-crm.vercel.app

Technical Skills

Programming	Python, TypeScript, JavaScript, SQL, C++, Dart
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Computer Vision, CNNs, Pandas, NumPy
Backend & Cloud	Node.js, NestJS, Express, AWS Lambda, REST APIs
Databases	PostgreSQL, MongoDB, DynamoDB, Prisma ORM
Infrastructure	Docker, Git, CI/CD pipelines, Linux (Ubuntu/WSL)

Languages

English	C1, fluent (Professional working proficiency)
German	A2, actively progressing toward B1
Urdu	Native

Siegen, October 5, 2026

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Signature